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### Conductive superlattice structures and methods of forming the same

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#### Abstract

A method of forming a metal superlattice structure includes depositing, on a substrate, a layer of a first metal with face-centered-cubic (fcc) crystal structure. The method further includes depositing a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal. The layer of the first metal may cause the layer of ruthenium metal to have fcc crystal structure.

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**Inventors:** Niimi; Hiroaki (Albany, NY), Leusink; Gerrit (Albany, NY), Maehara; Hiroki (Miyagi, JP), Abarra; Einstein Noel (Tokyo, JP), Watanabe; Naoki (Yamanashi, JP)

**Applicant:** Tokyo Electron Limited (Tokyo, JP)

**Family ID:** 1000008778685

**Assignee:** TOKYO ELECTRON LIMITED (Tokyo, JP)

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*Primary Examiner:* Yushin; Nikolay K

*Attorney, Agent or Firm:* Slater Matsil, LLP

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/233,683, filed on Aug. 16, 2021, which application is hereby incorporated herein by reference.

## TECHNICAL FIELD

(1) The present invention relates generally to semiconductor manufacturing, and, in particular embodiments, to methods for forming conductive superlattice structures and devices comprising such conductive superlattice structures.

## BACKGROUND

(2) As the semiconductor industry turns to alternative conductors to replace copper (Cu) metal for future interconnect nodes with improved device performance, much attention has been focused on evaluating the electrical performance of possible alternative metals or metal alloys. The crystal structure of a metal or a metal alloy is one of the most dominant factors that strongly affect the properties of an alloy, such as because the electronic structure changes drastically with the crystal structure. For ruthenium (Ru) metal, the most thermodynamically preferred crystal structure is hexagonal-closed-packed (hcp). However, relatively little attention has been paid to the face-centered-cubic (fcc) phase, which has been shown to nucleate in confined structures and may be present in tight-pitch interconnects.

## SUMMARY

(3) In accordance with an embodiment, a method of forming a metal superlattice structure includes: depositing, on a substrate, a layer of a first metal with face-centered-cubic (fcc) crystal structure; and depositing a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal.

(4) In accordance with another embodiment, a method of forming a metal superlattice structure includes: depositing, on a substrate including an active device, a first layer of a first metal with face-centered-cubic (fcc) crystal structure, the first layer being electrically coupled with the active device; depositing a first layer of ruthenium (Ru) metal on the first layer of the first metal, wherein the first layer of the first metal causes the first layer of Ru metal to have fcc crystal structure; depositing, on the first layer of Ru metal, a second layer of the first metal with fcc crystal structure; and depositing a second layer of Ru metal on the second layer of the first metal, wherein the second layer of the first metal causes the second layer of Ru metal to have fcc crystal structure.

(5) In accordance with yet another embodiment, a metal superlattice structure includes: a layer of a first metal with face-centered-cubic (fcc) crystal structure; and a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal.

(6) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosure, as claimed.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates a cross-sectional view of a superlattice structure, in accordance with some embodiments;

(3) FIG. 2 illustrates a cross-sectional view of a process chamber, in accordance with some embodiments;

(4) FIGS. 3 through 9 illustrate intermediate stages of manufacturing of a semiconductor structure, in accordance with some embodiments;

(5) FIGS. 10 through 13 illustrate intermediate stages of manufacturing of a semiconductor structure, in accordance with some embodiments;

(6) FIG. 14 is a flow chart of a method for a deposition process, in accordance with some embodiments; and

(7) FIG. 15 is a flow chart of another method for a deposition process, in accordance with some embodiments.

(8) Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale. The edges of features drawn in the figures do not necessarily indicate the termination of the extent of the feature.

#### DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(9) The making and using of various embodiments are discussed in detail below. It should be appreciated, however, that the various embodiments described herein are applicable in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use various embodiments, and should not be construed in a limited scope.

(10) Reference to “an embodiment” or “one embodiment” in the framework of the present description is intended to indicate that a particular configuration, structure, or characteristic described in relation to the embodiment is included in at least one embodiment. Hence, phrases such as “in an embodiment” or “in one embodiment” that may be present in one or more points of the present description do not necessarily refer to one and the same embodiment. Moreover, particular conformations, structures, or characteristics may be combined in any adequate way in one or more embodiments. The references used herein are provided merely for convenience and hence do not define the extent of protection or the scope of the embodiments.

(11) According to one or more embodiments of the present disclosure, this application relates to low electrical resistivity metal superlattice structures containing ruthenium metal and other low electrical resistivity transition metals and methods of forming the same. Superlattice structures containing ruthenium metal are attractive for use as interconnects in back-end-of-line (BEOL) applications and other advanced applications, such as for reasons of low electrical resistivity and small electron mean-free-path. Phonon-limited bulk resistivity of face-centered-cubic (fcc) ruthenium metal is less than about half that of the thermodynamically preferred hexagonal-closed-packed (hcp) ruthenium metal. This lower resistivity of the fcc ruthenium metal may be used to form superlattice structures including layers of ruthenium alternating other transition metals that have low electrical resistivity and fcc crystal structures. Further, these superlattice structures may be reasonably integrated with other materials due to favorable materials properties including a high melting point and formation of a smooth interface with other materials. In some examples, these superlattice structures may be integrated with dielectric materials. Diffusion barriers or liners may be added at the interfaces between the superlattice structures and the dielectric materials.

(12) According to some embodiments, a method is described for forming a low electrical resistivity metal superlattice structure. The method includes depositing a layer of a first metal with face-centered-cubic (fcc) crystal structure on a substrate and depositing a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal. In some embodiments, the face-centered-cubic (fcc) crystal structure of the layer of the first metal causes the layer of the ruthenium metal to have at least substantially fcc crystal structure.

(13) Embodiments of the disclosure are described in the context of the accompanying drawings. An embodiment of a conductive superlattice structure and methods of forming the same will be described using FIG. 1. An embodiment of a process chamber and methods of using the same for deposition processes will be described using FIG. 2. An embodiment of manufacturing a semiconductor structure including a conductive superlattice structure will be described using FIGS. 3 through 9. Another embodiment of manufacturing a semiconductor structure including a conductive superlattice structure will be described using FIGS. 10 through 13. Embodiments of methods for forming superlattice structures will be described using FIGS. 14 and 15.

(14) FIG. 1 illustrates a conductive superlattice structure **100**, in accordance with some embodiments. The conductive superlattice structure **100** includes alternating layers of ruthenium metal with fcc crystal structure and layers of a first metal with fcc crystal structure. In the example

of FIG. 1, five layers of ruthenium metal **102A**, **102B**, **102C**, **102D**, and **102E** and five layers of the first metal **101A**, **101B**, **101C**, **101D**, and **101E** are illustrated. However, any suitable number of alternating layers of ruthenium metal and the first metal may be formed and are within the scope of the disclosed embodiments. Thus, other conductive superlattice structures can contain one, two, three, or more alternating layers of ruthenium metal and the first metal. In one embodiment, as depicted in FIG. 1, a surface of each layer of ruthenium metal is in direct physical contact with a surface of a layer of the first metal.

(15) According to some embodiments, a method of forming the conductive superlattice structure **100** includes forming, on a substrate (not illustrated), a layer **101A** of a first metal with fcc crystal structure, and forming a layer **102A** of ruthenium metal with fcc crystal structure on the layer **101A** of the first metal. The method further includes depositing, on the layer **102A** of the ruthenium metal, a second layer **101B** of the first metal with fcc crystal structure, and depositing a second layer **102B** of ruthenium metal with fcc crystal structure on the second layer of the first metal. The alternating layers of the first metal and the ruthenium metal form a periodic conductive superlattice structure **100** that has lower electrical resistivity than a layer of hcp ruthenium metal with a same thickness as the conductive superlattice structure.

(16) According to some embodiments, a respective thickness of each layer of the ruthenium metal layers **102A**, **102B**, **102C**, **102D**, and **102E** is less than or equal to the c lattice parameter ( $c=0.428$  nm) of a hexagonal-closed-packed (hcp) ruthenium metal crystal structure. This thickness of each layer of ruthenium metal layer restricts formation of the thermodynamically preferred hcp ruthenium metal structure. As such, each layer of the ruthenium metal layers **102A**, **102B**, **102C**, **102D**, and **102E** has a fcc crystal structure.

(17) According to some embodiments, the composition of the first metal of the layers of the first metal **101A**, **101B**, **101C**, **101D**, and **101E** includes rhodium (Rh) metal, iridium (Ir) metal, palladium (Pd) metal, the like, or a combination thereof. For example, the first metal may include a mixture of palladium (Pd) metal and copper (Cu) metal.

(18) In one example, a layer **101A** of the first metal is deposited over a substrate (not illustrated) by physical vapor deposition (PVD) at a substrate temperature of about 200 Kelvin or less, and a layer **102A** of ruthenium metal is then deposited by physical vapor deposition at a substrate temperature of about 200 Kelvin or less. In one example, the substrate temperature is about 100 Kelvin. The physical vapor deposition may include alternately sputtering the first metal from a first target and sputtering the ruthenium metal from a second target. In one example, the physical vapor deposition is performed at a gas pressure of about 1 mTorr or less. According to some embodiments, a superlattice structure of one layer of fcc ruthenium metal (e.g., the layer **102A**) on one layer of fcc first metal (e.g., the layer **101A**) has lower electrical resistivity than a layer of hcp ruthenium metal with a same thickness as the superlattice structure.

(19) FIG. 2 illustrates a cross-sectional view of an example process chamber **150** for performing a deposition of a conductive superlattice structure **100** (see above, FIG. 1) over a substrate **50**, in accordance with some embodiments. The process chamber **150** may be configured for physical vapor deposition (PVD), although any suitable deposition method (e.g., ALD, PEALD, MBE (Molecular-beam epitaxy), or the like) may be used. The process chamber **150** includes a chuck **10** for holding a substrate **50**. The chuck **10** is configured to hold a substrate **50** (e.g., a semiconductor wafer that may have front end of the line (FEOL) structures including active devices formed on a top surface) inside the process chamber **150** to process a major surface of the substrate **50**. For example, the chuck **10** may be an electrostatic chuck including a bottom electrode or anode (not illustrated). In some embodiments, the chuck **10** is configured to rotate the substrate **50** during a deposition process, which may increase uniformity of deposition.

(20) The process chamber **150** is coupled to a vacuum pump **20** through an outlet **22**. The vacuum pump **20** is used to evacuate the process chamber **150** to a sufficiently low pressure for performing PVD, e.g. a pressure less than 1 mTorr. In some embodiments, the process chamber pressure

depends on the flow rate of the sputtering gas and is around, e.g., 0.5 mTorr. The vacuum pump **20** may be any suitable vacuum pump, e.g. a turbomolecular pump.

(21) A gas source **30** (e.g., a gas cylinder) is coupled to the process chamber **150** through an inlet **32**. The gas source **30** provides a sputtering gas to the process chamber **150**. The sputtering gas may be an inert gas such as argon, neon, krypton, xenon, carbon dioxide, the like, or a combination thereof. However, any suitable sputtering gas may be provided to the process chamber **150**.

(22) Two or more targets, such as target **70** and target **80**, are mounted on respective top cathodes **74** and **84** of the process chamber **150**. The target **70** and the target **80** are sources of the material to be deposited by physical vapor deposition over the substrate **50**. For example, in some embodiments, the target **70** is a sheet of the first metal (see above, FIG. 1) such as rhodium (Rh) metal, iridium (Ir) metal, palladium (Pd) metal, a mixture of palladium (Pd) metal and copper (Cu) metal, the like, or a combination thereof, and the target **80** is a sheet of ruthenium (Ru) metal. Although one target **70** of a first metal and one target **80** of ruthenium metal are illustrated in FIG. 2, any suitable number of targets may be present and mounted on respective cathodes.

(23) The substrate **50** is transported to the process chamber **150** and mounted on the chuck **10**. In some embodiments, the substrate **50** is cooled to a temperature below 200 K, such as 100 K, prior to being installed in the process chamber **150**. In some embodiments, the process chamber **150** is operated at a process temperature below 200 K, such as 100 K, during deposition processes.

(24) Once the substrate **50** is installed and the process chamber **150** is brought to a suitable pressure (e.g., less than 1 mTorr, such as around 0.5 mTorr) and a suitable cryogenic temperature (e.g., less than 200 K, such as in a range of 100 K to 130 K), the sputtering gas is flowed from the gas source **30** through the inlet **32** into the processing chamber **150**. In some embodiments, the sputtering gas is flowed at a rate in a range of 10 sccm to 100 sccm. The sputtering gas is excited to a plasma state to produce ions **40** (e.g., Ar.sup.+ ions). The ions **40** of the sputtering gas may be directed to bombard the target **70** or the target **80** by applying an electric field between the top cathodes **74** and **84** and the bottom electrode of the chuck **10**. In some embodiments, the electric field is generated with a DC power (as measured at the power source) in a range of 0.5 KW to 2 KW.

(25) The ions **40** bombard the target **70** and the target **80**, ejecting atoms **72** of the first metal from the target **70** and atoms **82** of ruthenium metal from the target **80**. The atoms **72** are deposited in a single fcc layer **101A** (see below, FIG. 3) of the first metal over the substrate **50**, and the atoms **82** of ruthenium metal are deposited in a single fcc layer **102A** (see below, FIG. 3) over the layer **101A** of the first metal. In some embodiments, alternating layers of the first metal **101A**, **101B**, **101C**, **101D**, and **101E** and layers of ruthenium metal **102A**, **102B**, **102C**, **102D**, and **102E** are formed by alternatively bombarding the target **70** and the target **80**. For example, while depositing the layer **101A**, the top cathode **74** may be activated and the top cathode **84** may be deactivated, resulting in the target **70** being bombarded and atoms **72** of the first metal being ejected while the target **80** is not bombarded. Likewise, while depositing the layer **102A**, the top cathode **84** may be activated and the top cathode **74** may be deactivated, resulting in the target **80** being bombarded and atoms **82** of ruthenium metal being ejected while the target **70** is not bombarded. The deposition of each layer may be performed for a duration in a range of 5 seconds to 30 seconds, which may occur over one rotation of the substrate **50** by the chuck **10**.

(26) In some embodiments, the target **70** and the target **80** are bombarded at the same time during deposition while the chuck **10** rotates the substrate **50** once to form a monolayer including atoms **72** of the first metal from the target **70** and atoms **82** of ruthenium metal from the target **80**. In some embodiments, a single layer **101A** of the first metal is deposited with an fcc structure by bombarding the target **70** prior to forming a subsequent layer or layers by bombarding the target **70** and the target **80** at the same time.

(27) FIGS. 3 through 9 illustrate intermediate stages of manufacturing of a semiconductor structure **200** in a cross-sectional view, in accordance with some embodiments. FIGS. 3 through 5 illustrate the formation of a conductive superlattice structure **100** over a substrate **50**. FIGS. 6 through 9

illustrate a subtractive patterning of the conductive superlattice structure **100** and a subsequent formation of an interconnect layer over the patterned conductive superlattice structure **100**.

(28) FIG. 3 illustrates the formation of a layer **101A** of a first metal over an example substrate **50** and the formation of a layer **102A** of ruthenium metal over the first layer **101A**, in accordance with some embodiments. In various embodiments, the substrate **50** may be a part of, or include, a semiconductor device (e.g. one or more transistors), and may have undergone a number of steps of processing following, for example, a conventional process. The substrate **50** accordingly may comprise layers of semiconductors useful in various microelectronics. For example, the semiconductor structure **200** may comprise the substrate **50** in which various device regions are formed.

(29) As illustrated in FIG. 3, the example substrate **50** includes a semiconductor substrate **225**, a shallow trench isolation (STI) region **220**, a metal gate **210**, a high-k gate dielectric **214**, a pair of source/drain spacers **212**, a source/drain contact etch-stop layer (CESL) **218**, a first interlayer dielectric (ILD) **230**, a second ILD **232**, and a contact **235**.

(30) In various embodiments, the semiconductor substrate **225** is a silicon wafer (e.g., a bulk crystalline silicon wafer), or a silicon-on-insulator (SOI) wafer. In certain embodiments, the semiconductor substrate **225** comprises silicon germanium, silicon carbide, gallium arsenide, gallium nitride, or other compound semiconductors. In other embodiments, the semiconductor substrate **225** comprises heterogeneous layers such as silicon germanium on silicon, gallium nitride on silicon, silicon carbon on silicon, as well layers of silicon on a silicon or SOI substrate. In various embodiments, the semiconductor substrate **225** is patterned or embedded in other components of the semiconductor device.

(31) A gate structure of a transistor (e.g., a FinFET) is over the semiconductor substrate **225**. The gate structure comprises a metal gate **210** (e.g., a multilayer metal stack comprising Ta, TaN, TiN, W, the like, or a combination thereof) and a gate dielectric **214** (e.g., a high-k gate dielectric comprising  $\text{HfO}_2$ ,  $\text{Al}_2\text{O}_3$ , the like, or a combination thereof) inlaid within a recess formed earlier between a pair of source/drain spacers **212** (e.g.,  $\text{SiO}_2$  spacers). As illustrated in FIG. 3, the metal gate **210** and the gate dielectric **214** are a portion of the metal gate structure and extend over a shallow trench isolation (STI) region **220** in recesses between semiconductor fins formed earlier, for example, by etching the semiconductor substrate **225**. The semiconductor fins are not illustrated, being located along planes parallel to the plane of the cross-sectional view in FIG. 3. Source/drain regions (not illustrated) may be present in or on portions of the semiconductor fins adjacent to the gate structure.

(32) The first ILD **230** is over the STI region **220** and the semiconductor substrate **225** and is adjacent to sidewalls of the source/drain spacers **212**. The first ILD **230** comprises one or more insulators such as  $\text{SiO}_2$  or a silicon oxide based low-k dielectric (e.g., porous oxides, fluorosilicate glass (FSG), and organosilicate glass (OSG)). In some embodiments, a source/drain contact etch-stop layer (CESL) **218** (e.g., a  $\text{Si}_3\text{N}_4$  layer) lines the bottom surface of the first ILD **230**.

(33) A second ILD **232** is over the first ILD **230**, the gate structure (including the metal gate **210** and the gate dielectric **214**), and the source/drain spacers **212**. The second ILD **232** may include similar materials as the first ILD **230** (see above). In some embodiments, the second ILD **232** includes a bottom layer that is an etch stop layer (ESL) that comprises a dielectric such as  $\text{Si}_3\text{N}_4$ ,  $\text{SiO}_2$ , SiC, or SiCN (not shown). A contact **235** extends through the second ILD **232** and physically and electrically couples with the metal gate **210**. Additionally, in some embodiments, a conductive ESL (not shown) comprising, for example, TiN or TaN is formed over the second ILD **232** and the contact **235** before subsequent layers (e.g., the layers **101A** and **102A**) are deposited.

(34) The substrate **50** is a non-limiting example and is described above for illustrative purposes. Any suitable substrate **50** with any suitable composition or arrangement of features may be used for

the manufacturing of a semiconductor structure **200** that includes a conductive superlattice structure **100** formed on the substrate **50**, and should be understood to be within the scope of the disclosed embodiments.

(35) A layer **101A** is formed from a first metal over the substrate **50**. In some embodiments, the layer **101A** is formed over the second ILD **232** and the contact **235** in the example illustrated by FIG. 3. In other embodiments, the second ILD **232** and the contact **235** are not present and the layer **101A** is formed directly on the gate structure including the metal gate **210** and the gate dielectric **214**. The layer **101A** may be formed over any suitable substrate **50**, and all such substrates are within the scope of the disclosed embodiments.

(36) The composition of the first metal includes rhodium (Rh) metal, iridium (Ir) metal, palladium (Pd) metal, or the like, or a combination thereof. In another embodiment, the first metal includes a mixture of palladium (Pd) metal and copper (Cu) metal. Using a first metal including one or more metals from the group of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd) metal may result in the layer **101A** having a fcc structure and may further restrict formation of hcp structured ruthenium metal in a subsequent deposition of ruthenium (see below). The layer **101A** may have a thickness less than 0.428 nm, such as about 0.4 nm, which is less than the center spacing of fcc structured ruthenium metal. For example, the layer **101A** may be an atomic monolayer of the first metal. The layer **101A** may be formed using physical vapor deposition in a process chamber **150** as described above with respect to FIG. 2. However, any suitable method may be used to form the layer **101A**.

(37) Next, a layer **102A** of ruthenium metal is formed over the layer **101A** of the first metal. The layer **102A** may have a thickness less than 0.428 nm, which is less than the center spacing of fcc structured ruthenium metal. For example, the layer **102A** may be an atomic monolayer of ruthenium metal. The lattice constant of the layer **102A** of fcc structured ruthenium metal may be the range of 0.382 nm to 0.383 nm.

(38) Forming the layer **102A** of ruthenium metal over the layer **101A** of the first metal, which includes one or more metals from the group of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd), may restrict the formation of otherwise thermodynamically favored hcp structured ruthenium and result in the layer **102A** having an fcc structure. This may lead to the conductive superlattice structure **100** having lower resistivity, due to the lower resistivity of fcc structured ruthenium in comparison with hcp structured ruthenium or fcc structured copper. Ruthenium with an fcc structure is a promising metal for narrow interconnects with high conductivity and low resistivity due to having a small product of its bulk resistivity ( $\rho_{\text{sub.o}}$ ) times its bulk electron free path ( $\lambda$ ),  $\rho_{\text{sub.o}} \times \lambda$ . For example, the  $\rho_{\text{sub.o}} \times \lambda$  of fcc structured ruthenium is  $2.47 \times 10^{-16} \Omega\text{-m}$ , which is smaller than the  $\rho_{\text{sub.o}} \times \lambda$  of fcc structured ruthenium ( $6.44 \times 10^{-16} \Omega\text{-m}$  or  $4.76 \times 10^{-16} \Omega\text{-m}$ ) or the  $\rho_{\text{sub.o}} \times \lambda$  of fcc structured copper ( $6.70 \times 10^{-16} \Omega\text{-m}$ ). The layer **102A** may be formed using physical vapor deposition in a process chamber **150** as described above with respect to FIG. 2. However, any suitable method may be used to form the layer **102A**.

(39) Next, in FIG. 4, additional alternating layers of a first metal **101B**, **101C**, **101D**, and **101E** with fcc crystal structure and layers of ruthenium metal **102B**, **102C**, **102D**, and **102E** with fcc crystal structure are formed. In some embodiments, each additional layer of the first metal or of ruthenium is formed with a physical vapor deposition process as described above with respect to FIG. 2.

(40) In the example of FIG. 4, the conductive superlattice structure **100** includes five layers of the first metal and five layers of ruthenium metal. However, any suitable number of alternating layers of ruthenium metal and the first metal may be formed and are within the scope of the disclosed embodiments.

(41) In FIG. 5, a capping layer **104** is formed over the conductive superlattice structure **100**. The capping layer **104** may be or include a protection layer to protect the conductive superlattice structure **100** from oxidation or a chemical mechanical polish (CMP) stop layer for stopping a

subsequent CMP (see below, FIG. 8). In some embodiments where the capping layer **104** is or includes a protection layer, the capping layer comprises titanium nitride (TiN) and is formed using PVD in the same process chamber **150** after forming the last layer of the conductive superlattice structure **100**. In some embodiments where the capping layer **104** is or includes a CMP stop layer, the capping layer comprises silicon nitride (SiN) and is formed using PVD in the same process chamber **150** after forming the last layer of the conductive superlattice structure **100**. However, any suitable process and materials may be used to form the capping layer **104**. In some embodiments, the capping layer **104** is omitted.

(42) In FIG. 6, the conductive superlattice structure **100** is patterned to form conductive features **110**, in accordance with some embodiments. The conductive features **110** (e.g., conductive lines) may form electrical connections in an interconnect structure, such as for back end of the line (BEOL) applications. In some embodiments, the conductive features **110** have widths (also referred to as critical dimensions) in a range of 5 nm to 15 nm, or less than 10 nm.

(43) As an example of forming the conductive features **110**, a patterned photoresist layer (not illustrated) is formed over the top surface of the conductive superlattice structure **100** using a photomask with a pattern designed for, e.g., conductive lines. Next, openings **248** are etched using the patterned photoresist layer as an etch mask. The etching may be performed using a suitable anisotropic etch technique, for example, a reactive ion etch (RIE) process using fluorine chemistry. However, any suitable etching process may be used. The etch may be chosen to be selective with the material underlying the conductive features **110**, e.g. dielectric material the second ILD **232**. In some embodiments, the conductive superlattice structure **100** is formed over another conductive material (e.g., a metal such as tungsten, cobalt, or copper, such as in the contact **235**), and the etch process is selective with the conductive material underlying the conductive superlattice structure **100**.

(44) In FIG. 7, a first intermetal dielectric (IMD) **250** is formed over the conductive features **110** and fills the openings **248** (see above, FIG. 6), in accordance with some embodiments. The IMD **250** comprises an insulating material such as SiO<sub>2</sub> or a silicon oxide based low-k dielectric (e.g., porous oxides, fluorosilicate glass (FSG), and orthosilicate glass (OSG)), similar to the materials of the first ILD **230** and the second ILD **232** described above with respect to FIG. 3. The IMD **250** may be formed with a suitable process such as CVD or the like.

(45) In FIG. 8, an excess portion of the IMD **250** is removed with a suitable etchback technique, for example, a chemical mechanical polish (CMP). In some embodiments, respective top portions of the capping layer **104** (if present) or respective top portions of the conductive features **110** are also removed.

(46) In FIG. 9, an interconnect layer is formed over the conductive features **110** and the IMD **250**, in accordance with some embodiments. The interconnect layer includes an IMD **260** with a conductive via **262** and a conductive line **266** formed therein. The IMD **260** may be formed using similar methods and materials as described above for the IMD **250** with respect to FIG. 11. The conductive via **262** connects the conductive line **266** to a portion of the conductive feature **110** disposed directly below the conductive line **266**. The conductive via **262** and the conductive line **266** may be formed with, for example, a conventional dual-damascene process using a conductive material such as copper or the like. As known by a person skilled in the art, the dual-damascene flow comprises patterning openings (e.g., holes for conductive vias **262** and trenches for conductive lines **266**) in the IMD **260** using a via-first or a trench-first patterning sequence, depositing a conformal barrier metal (e.g., TiN or TaN) liner, filling the openings with metal (e.g., using Cu electroplating), and removing all excess conductive material from the top surface of the IMD **260** using a planarization process such as chemical mechanical planarization (CMP), thereby forming the conductive vias **262** and conductive lines **266** inlaid in the IMD **260**.

(47) The interconnect layer including the IMD **260** with the conductive via **262** and the conductive line **266** is included as a non-limiting example. Any suitable interconnect layer may be formed over

the conductive features **110** and IMD **250** and is within the scope of the disclosed embodiments. For example, an additional conductive superlattice structure similar to the conductive superlattice structure **100** (see above, FIGS. **3-4**) may be formed over the conductive features **110** and IMD **250** and subsequently patterned to form additional conductive features, with spaces between the conductive features subsequently filled by an IMD.

(48) FIGS. **10** through **13** illustrate intermediate stages of manufacturing of another semiconductor structure **300** in a cross-sectional view, in accordance with some embodiments. The semiconductor structure **300** includes a conductive superlattice structure **400** that is formed and patterned with a semi-damascene process.

(49) FIG. **10** illustrates a substrate **350** that is similar to the substrate **50** as described above with respect to FIG. **3** but without the contact **235** formed through the second ILD **232**. An opening **234** is formed through the second ILD **232** to expose a top surface of the metal gate **210**. The opening **234** may be formed with a suitable lithographic patterning process. In some embodiments, the opening **234** has tapered sidewalls and a trapezoidal profile in a cross-sectional view.

(50) In FIG. **11**, a conductive superlattice structure **400** is formed over the second ILD **232** and fills the opening **234** (see above, FIG. **10**), physically contacting the top surface of the metal gate **210**. The conductive superlattice structure **400** may be formed using similar methods and materials as the conductive superlattice structure **100** as described above with respect to FIGS. **3-4**. As illustrated in FIG. **11**, lower layers **101A**, **101B**, **102A**, and **102B** fill the opening **234**. However, any suitable number of layers of the conductive superlattice structure **400** may be deposited in and fill the opening **234**.

(51) In FIG. **12**, the conductive superlattice structure **400** is patterned to form conductive features **410** separated by air gaps **252**. The conductive superlattice structure **400** may be patterned by a similar process as the conductive superlattice structure **100** as described above with respect to FIG. **6**, and the details are not repeated herein. As illustrated in FIG. **12**, a conductive feature **410** may include a lower via portion extending through the second ILD **232** to contact the metal gate **210** and an upper line portion above the second ILD **232**. As such, the conductive features **410** are formed with a semi-damascene process.

(52) In FIG. **13**, an interconnect layer is formed over the conductive features **410** and the air gaps **252**, in accordance with some embodiments. The interconnect layer includes an IMD **260** with a conductive via **262** and a conductive line **266** formed therein. The conductive via **262** connects the conductive line **266** to a portion of the conductive feature **410** disposed directly below the conductive line **266**. The IMD **260**, conductive via **262**, and conductive line **266** may be formed using similar methods and materials as described above with respect to FIG. **9**, and the details are not repeated herein.

(53) The air gaps **252** may be sealed by the IMD **260** and may contain air or other gases. The air gaps **252** being disposed between adjacent conductive features **410** that include alternating layers of ruthenium metal and first metal with fcc structure may reduce the resistance of the conductive features **410**, thus increasing device performance.

(54) FIG. **14** illustrates a flow chart diagram of a method **500** for a deposition process, in accordance with some embodiments. In step **502**, a first layer **101A** of a first metal with fcc crystal structure is deposited on a substrate **50**, as described above with respect to FIGS. **2** and **3**. In step **504**, a first layer **102A** of ruthenium metal with fcc crystal structure is deposited on the first layer **101A** of the first metal, as described above with respect to FIGS. **2** and **3**.

(55) FIG. **15** illustrates a flow chart diagram of a method **600** for a deposition process, in accordance with some embodiments. In step **602**, a first layer **101A** of a first metal with fcc crystal structure is deposited on a substrate **50**, as described above with respect to FIGS. **2** and **3**. In step **604**, a first layer **102A** of ruthenium metal is deposited on the first layer **101A** of the first metal, so that the first layer **101A** of the first metal causes the first layer **102A** of ruthenium to have fcc crystal structure, as described above with respect to FIGS. **2** and **3**. In step **606**, a second layer

**101B** of the first metal with fcc crystal structure is deposited on the first layer **102A** of ruthenium metal, as described above with respect to FIGS. 2 and 4. In step **608**, a second layer **102B** of ruthenium metal is deposited on the second layer **101B** of the first metal, so that the second layer **101B** of the first metal causes the second layer **102B** of ruthenium to have fcc crystal structure, as described above with respect to FIGS. 2 and 4.

(56) Example embodiments of the disclosure are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.

(57) Example 1. A method of forming a metal superlattice structure, the method including: depositing, on a substrate, a layer of a first metal with face-centered-cubic (fcc) crystal structure; and depositing a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal.

(58) Example 2. The method of example 1, where the substrate includes a contact and a transistor, the contact being electrically coupled with the transistor, the metal superlattice structure being formed to electrically couple the metal superlattice structure to the transistor through the contact.

(59) Example 3. The method of one of examples 1 or 2, further including: forming a conductive line by patterning the layer of the first metal and the layer of Ru metal; and forming an interconnect layer over the conductive line, where a conductive feature of the interconnect layer is coupled with the conductive line.

(60) Example 4. The method of one of examples 1 to 3, where a surface of the layer of Ru metal is in direct physical contact with a surface of the layer of the first metal.

(61) Example 5. The method of one of examples 1 to 4, where a thickness of the layer of Ru metal is less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.

(62) Example 6. The method of example 5, where the thickness of the layer of Ru metal restricts formation of the hcp Ru metal structure.

(63) Example 7. The method of one of examples 1 to 6, where the layer of the first metal is selected from the group consisting of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd) metal.

(64) Example 8. The method of one of examples 1 to 7, where the metal superlattice structure has a lower electrical resistivity than a layer of hcp Ru metal with a same thickness as the metal superlattice structure.

(65) Example 9. The method of one of examples 1 to 8, further including: depositing, on the layer of Ru metal, a second layer of the first metal with fcc crystal structure; and depositing a second layer of Ru metal with fcc crystal structure on the second layer of the first metal.

(66) Example 10. The method of example 9, where the alternating layers of the first metal and the Ru metal form a periodic structure that has lower electrical resistivity than a layer of hcp Ru metal with a same thickness as the metal superlattice structure.

(67) Example 11. A method of forming a metal superlattice structure, including: depositing, on a substrate including an active device, a first layer of a first metal with face-centered-cubic (fcc) crystal structure, the first layer being electrically coupled with the active device; depositing a first layer of ruthenium (Ru) metal on the first layer of the first metal, where the first layer of the first metal causes the first layer of Ru metal to have fcc crystal structure; depositing, on the first layer of Ru metal, a second layer of the first metal with fcc crystal structure; and depositing a second layer of Ru metal on the second layer of the first metal, where the second layer of the first metal causes the second layer of Ru metal to have fcc crystal structure.

(68) Example 12. The method of example 11, further including: forming a first conductive feature by patterning the second layer of Ru metal, the second layer of the first metal, the first layer of Ru metal, and the first layer of the first metal; depositing an intermetal dielectric over the first conductive feature; and forming a second conductive feature in the intermetal dielectric, where the second conductive feature is coupled with the first conductive feature.

(69) Example 13. The method of one of examples 11 or 12, where thicknesses of each of the first

layer of Ru metal and the second layer of Ru metal are less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.

(70) Example 14. The method of one of examples 11 to 13, where the first layer of the first metal, the second layer of the first metal, the first layer of Ru metal, and the second layer of Ru metal are deposited by physical vapor deposition at a substrate temperature of less than 200 Kelvin.

(71) Example 15. A metal superlattice structure, including: a layer of a first metal with face-centered-cubic (fcc) crystal structure; and a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal.

(72) Example 16. The structure of example 15, where a surface of the layer of Ru metal is in direct physical contact with a surface of the layer of the first metal.

(73) Example 17. The structure of one of examples 15 or 16, where a thickness of the layer of Ru metal is less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.

(74) Example 18. The structure of one of examples 15 to 17, where a thickness of the layer of Ru metal restricts formation of an hcp Ru metal structure.

(75) Example 19. The structure of one of examples 15 to 18, where the layer of the first metal is selected from the group consisting of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd) metal.

(76) Example 20. The structure of one of examples 15 to 18, where the layer of the first metal includes a mixture of palladium (Pd) metal and copper (Cu) metal.

(77) Although the description has been described in detail, it should be understood that various changes, substitutions, and alterations may be made without departing from the spirit and scope of this disclosure as defined by the appended claims. The same elements are designated with the same reference numbers in the various figures. Moreover, the scope of the disclosure is not intended to be limited to the particular embodiments described herein, as one of ordinary skill in the art will readily appreciate from this disclosure that processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, may perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

## Claims

1. A method of forming a metal superlattice structure, the method comprising: depositing, on a substrate comprising an active device, a layer of a first metal with face-centered-cubic (fcc) crystal structure, the layer of the first metal being electrically coupled with the active device; and depositing a layer of ruthenium (Ru) metal with fcc crystal structure on the layer of the first metal.
2. The method of claim 1, wherein the substrate further comprises a contact and wherein the active device comprises a transistor, the contact being electrically coupled with the transistor, the metal superlattice structure being formed to electrically couple the metal superlattice structure to the transistor through the contact.
3. The method of claim 1, further comprising: forming a conductive line by patterning the layer of the first metal and the layer of Ru metal; and forming an interconnect layer over the conductive line, wherein a conductive feature of the interconnect layer is coupled with the conductive line.
4. The method of claim 1, wherein a surface of the layer of Ru metal is in direct physical contact with a surface of the layer of the first metal.
5. The method of claim 1, wherein a thickness of the layer of Ru metal is less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.
6. The method of claim 5, wherein the thickness of the layer of Ru metal restricts formation of the

hcp Ru metal crystal structure.

7. The method of claim 1, wherein the layer of the first metal is selected from the group consisting of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd) metal.

8. The method of claim 1, wherein the metal superlattice structure has a lower electrical resistivity than a layer of hcp Ru metal with a same thickness as the metal superlattice structure.

9. The method of claim 1, further comprising: depositing, on the layer of Ru metal, a second layer of the first metal with fcc crystal structure; and depositing a second layer of Ru metal with fcc crystal structure on the second layer of the first metal.

10. The method of claim 9, wherein the alternating layers of the first metal and the Ru metal form a periodic structure that has lower electrical resistivity than a layer of hcp Ru metal with a same thickness as the metal superlattice structure.

11. A method of forming a metal superlattice structure, comprising: depositing, on a substrate comprising an active device, a first layer of a first metal with face-centered-cubic (fcc) crystal structure, the first layer being electrically coupled with the active device; depositing a first layer of ruthenium (Ru) metal on the first layer of the first metal, wherein the first layer of the first metal causes the first layer of Ru metal to have fcc crystal structure; depositing, on the first layer of Ru metal, a second layer of the first metal with fcc crystal structure; and depositing a second layer of Ru metal on the second layer of the first metal, wherein the second layer of the first metal causes the second layer of Ru metal to have fcc crystal structure.

12. The method of claim 11, further comprising: forming a first conductive feature by patterning the second layer of Ru metal, the second layer of the first metal, the first layer of Ru metal, and the first layer of the first metal; depositing an intermetal dielectric over the first conductive feature; and forming a second conductive feature in the intermetal dielectric, wherein the second conductive feature is coupled with the first conductive feature.

13. The method of claim 11, wherein thicknesses of each of the first layer of Ru metal and the second layer of Ru metal are less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.

14. The method of claim 11, wherein the first layer of the first metal, the second layer of the first metal, the first layer of Ru metal, and the second layer of Ru metal are deposited by physical vapor deposition at a substrate temperature of less than 200 Kelvin.

15. A metal superlattice structure, comprising: a first layer of a first metal with face-centered-cubic (fcc) crystal structure; a first layer of ruthenium (Ru) metal with fcc crystal structure on the first layer of the first metal; a second layer of the first metal with face-centered-cubic (fcc) crystal structure on the first layer of the Ru metal; and a second layer of Ru metal with fcc crystal structure on the second layer of the first metal.

16. The structure of claim 15, wherein a surface of the first layer of Ru metal is in direct physical contact with a surface of the first layer of the first metal.

17. The structure of claim 15, wherein a thickness of the first layer of Ru metal is less than or equal to the c lattice parameter of hexagonal-closed-packed (hcp) Ru metal crystal structure.

18. The structure of claim 15, wherein a thickness of the first layer of Ru metal restricts formation of an hcp Ru metal crystal structure.

19. The structure of claim 15, wherein the first metal is selected from the group consisting of rhodium (Rh) metal, iridium (Ir) metal, and palladium (Pd) metal.

20. The structure of claim 15, wherein the first layer of the first metal includes a mixture of palladium (Pd) metal and copper (Cu) metal.

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